THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The line $4x + 2y = 9$ is parallel to the line $kx + 4y = 11$. The value of k is: 1

(a) 8 **(b)** 9 **(c)** 10 **(d)** 7

2. The radii of two spheres are in the ratio $1 : 6$. The ratio of their volumes is: 1

(a) $1 : 6$ **(b)** $1 : 216$ **(c)** $1 : 210$ **(d)** $1 : 36$

3. The point on the x-axis equidistant from $(1, 5)$ and $(5, -5)$ is: 1

(a) $(-3, 0)$ **(b)** $(3, 0)$ **(c)** $(4, 0)$ **(d)** $(0, 3)$

4. The point $(k, 31)$ lies on $y = 7x + 3$. Then $k =$ 1

(a) 6 **(b)** 5 **(c)** 3 **(d)** 4

5. The length of the repeating block in the decimal expansion of $\frac{1}{7}$ is: 1

(a) 5 **(b)** 8 **(c)** 6 **(d)** 7

6. If $x + \frac{1}{x} = 3$, then $x^2 + \frac{1}{x^2} =$ 1

(a) 7 **(b)** 9 **(c)** 6 **(d)** 8

7. The angles of a cyclic quadrilateral $ABCD$ at A and C are in the ratio $2 : 3$. Then $\angle A =$ 1

(a) 92° **(b)** 62° **(c)** 82° **(d)** 72°

8. A square has area 169 cm^2. Its perimeter (cm) is: 1

(a) 53 **(b)** 54 **(c)** 52 **(d)** 51

9. A bag has 3 red and 5 blue balls. After adding 4 more red balls, the probability of drawing a red ball is: 1

(a) $\frac{7}{12}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{5}{12}$ **(d)** $\frac{7}{13}$

10. The sum of the first 13 even natural numbers is: 1

(a) 182 **(b)** 195 **(c)** 208 **(d)** 169

11. The number of factors of 72 is: 1

(a) 13 **(b)** 12 **(c)** 14 **(d)** 11

12. A group of 12 people has average weight 71 kg. Another group has average weight 74 kg. Together the average weight is 73 kg. The number of people in the second group is: 1

(a) 24 **(b)** 26 **(c)** 23 **(d)** 25

13. Set s = 0. While n > 0: add (n mod 10) to s, then replace n by the quotient of n divided by 10. Starting with n = 32481, the final s is: 1

(a) 17 **(b)** 20 **(c)** 19 **(d)** 18

14. The diagonals of a rhombus are 10 cm and 24 cm. Its perimeter is: 1

(a) 56 cm **(b)** 78 cm **(c)** 52 cm **(d)** 65 cm

15. The line $3x + 2y = 6$ cuts the axes at two points. The area of the triangle it forms with the axes (square units) is: 1

(a) 5 **(b)** 2 **(c)** 4 **(d)** 3

16. Take $\pi = \frac{22}{7}$. The total surface area of a cone of radius 10.5 cm and slant height 17.5 cm is: 1

(a) 1017 cm^2 **(b)** 831 cm^2 **(c)** 924 cm^2 **(d)** 1110 cm^2

17. If $ab > 0$ and $a + b < 0$, the point (a, b) lies in: 1

(a) Quadrant III **(b)** Quadrant IV **(c)** Quadrant I **(d)** Quadrant II

18. If $p(x) = 5x + 2$ and $q(x) = 3x - 7$, then $p(x) + q(x) =$ 1

(a) $8x - 5$ **(b)** $8x - 6$ **(c)** $8x - 4$ **(d)** $9x - 5$

19. $\sqrt{3} \times \sqrt{12} =$ 1

(a) 8 **(b)** 5 **(c)** 7 **(d)** 6

20. The value of 1003^2 is: 1

(a) 1006009 **(b)** 1006011 **(c)** 1006008 **(d)** 1006010

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. A taxi fare is ₹82 for 3 km and ₹163 for 12 km. Assuming the fare is linear in the distance, find the fare per km, the fixed charge and the fare for 23 km. 2

22. Take $\pi = \frac{22}{7}$. A cylinder of height 6 cm has volume 2079 cm^3. Find its radius and its curved surface area. 2

23. Find the midpoint of the segment joining $(3, -4)$ and $(6, 7)$. 2

24. For $p(x) = 5x + 1$ and $q(x) = 2x - 8$, find the zero of $p(x) - q(x)$. 2

25. Find the decimal expansion of $\frac{5}{22}$. 2

26. If $x + y = 11$ and $xy = 28$, find $x^2 + y^2$. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Express $1.2\overline{3}$ in the form $\frac{p}{q}$. 3

28. If $a - b = 4$ and $ab = 21$, find $a^2 + b^2$ and $(a + b)^2$. 3

29. In a cyclic quadrilateral $ABCD$, $\angle A = (3x + 20)^\circ$ and $\angle C = (2x - 10)^\circ$. Find x, $\angle A$ and $\angle C$. 3

30. A rectangular sheet 30 cm $\times$ 14 cm has a semicircle of diameter 14 cm added on one breadth (take $\pi = \frac{22}{7}$). Find the perimeter and the area of the whole figure. 3

31. In a survey of 200 people, 70 like tea only, 50 like coffee only and 30 like both. Find the probability that a person chosen at random likes tea (tea only or both), and the probability that the person likes neither (the rest). 3

OR

The 3rd term of an AP is 16 and the 8th term is 41. Find the AP and its 20th term. 3

32. Show that the statement "A number divisible by 3 and by 9 must be divisible by 27" is false by giving a counterexample. 3

OR

The average score of 45 students in a test is 80. When 54 more students take the test, the average score of all 99 students becomes 68. Find the average score of the 54 additional students. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Two chords of the same circle have lengths 10 cm and 24 cm and the circle has radius 13 cm. Find the distances of the chords from the centre and say which chord is nearer to the centre. 5

34. A triangular field has sides 13 m, 13 m and 10 m. Find its area, and the cost of ploughing it at ₹5 per m^2. 5

35. A factory tested 200 bulbs and found 8 defective. Find the probability that a bulb is defective, and estimate the number of defective bulbs in a lot of 5000. 5

36. A sum of ₹10000 is invested at 10% per year compound interest. Show that the amounts at the end of years form a GP and find the amount after 3 years. 5

OR

Take the two-digit number 47. Subtract the sum of its digits and show that the result is divisible by 9. Prove the same for any two-digit number. 5

37. 510 mL of a 22% sugar solution is mixed with 680 mL of a 15% sugar solution. Find the percentage concentration of the mixture. 5

OR

Start with n = 9. Repeat: if n is even replace it by n/2, otherwise by 3n + 1, until n = 1. Write the sequence and the number of steps. 5

38. The perimeter of $\triangle ABC$ is 82 cm. D, E, F are the midpoints of its sides. Find the perimeter of $\triangle DEF$ and the length of each of its sides in terms of the sides of $\triangle ABC$. 5

OR

A line has slope 5 and passes through $(4, 25)$. Find its equation in the form $y = mx + c$, its x-intercept and the value of y when $x = 2$. 5

END OF QUESTION PAPER — Total: 80 Marks